Car Market App: Folder & File Guide

Har folder aur file mein kya aayega, aur uska use kya hai | Flutter MVVM structure

Is doc ko reference ki tarah use karo. Koi bhi nayi file banate time yahan dekho ki wo kahan jaayegi.

MVVM ka basic idea

Project teen layers mein bata hua hai. Har layer ka ek hi kaam hai.

Layer	Folder	Kaam
View	`features/*/view/`	Sirf UI dikhana. Koi business logic nahi.
ViewModel	`features/*/viewmodel/`	Screen ki state + logic (`ChangeNotifier`). Repository se data leta hai.
Model	`data/models/`	Data ka shape (Car, Offer, Order...).
Repository	`data/repositories/`	ViewModel aur backend ke beech ki deewar. Backend badle toh sirf yahi badlega.
Service	`data/services/`	Actual Firebase/API calls.

Data ka flow:

`View` -> `ViewModel` -> `Repository` -> `Service` -> Firebase

Rule: koi layer apne se do level neeche seedha baat nahi karti. View kabhi Repository ya Firebase ko nahi chhuta.

Root level files

File / Folder	Kya aayega	Use
`pubspec.yaml`	Packages (provider, go_router, google_fonts...), assets ki list, app ka naam/version	Project ka config. Naya package yahin add hota hai (`flutter pub add`).

File / Folder	Kya aayega	Use
`analysis_options.yaml`	Lint rules (flutter_lints)	Code quality warnings. Default rakh sakte ho.
`README.md`	Setup steps, project overview, screenshots	Dusre log (ya recruiters) ko project samajhane ke liye.
`.gitignore`	`build/`, `.dart_tool/`, `google-services.json` jaisi files	Git mein kya nahi jaana chahiye. Flutter khud bana deta hai.
`assets/images/`	Car images, logos, illustrations (PNG/JPG)	App ke andar bundle hone wali images. Path `pubspec.yaml` mein declare hona chahiye.
`assets/icons/`	SVG/PNG icons (payment logos, bank icons)	Custom icons jo Material icons mein nahi hain.
`android/`, `ios/`	Platform-specific code, permissions, `google-services.json`	Flutter khud banata hai. Bas Firebase config aur permissions yahan jaate hain.
`test/`	Unit tests aur widget tests	ViewModels ka test yahan likho (total calculation, offer logic).

lib/main.dart

File	Kya aayega	Use
`main.dart`	`runApp()`, `Firebase.initializeApp()`, `MultiProvider` mein saare ViewModels, `MaterialApp.router` with theme aur router	App ka entry point. Yahan sab kuch wire hota hai, par koi UI ya logic nahi likhna.

lib/core/

Ye folder poore app mein share hone wali cheezein rakhta hai. Isme koi feature-specific cheez nahi aani chahiye.

core/theme/

File	Kya aayega	Use
`app_theme.dart`	`AppColors` class (bg, card, border, muted text, success, error), `ThemeData` dark theme, font (Urbanist), button/appbar/input styles	Poore app ka look ek jagah se control hota hai. Color badalna ho toh sirf yahin badlo.

core/router/

File	Kya aayega	Use
`app_router.dart`	`GoRouter` config, saare route paths, bottom nav ke liye `ShellRoute`, redirect logic (login nahi hai toh login screen)	Screens ke beech navigation. Route name/path ek hi jagah defined rahe.

core/widgets/

Reusable widgets jo 2 ya zyada features mein use hote hain.

File	Kya aayega	Use
`num_pad.dart`	3x4 numeric keypad, `onDigit` aur `onBackspace` callbacks	Make Offer, Enter PIN, Top Up teeno screens mein same widget.
`pin_dots.dart`	4 dots jo digit enter hone pe fill hote hain	PIN screens (payment aur top up).
`primary_button.dart`	Full-width white rounded button, loading state	Har screen ka main action button (Continue, Apply, Confirm).
`app_card.dart` (optional)	Rounded dark container with border	Order card, address card, payment method row.
`status_popup.dart` (optional)	Icon + title + subtitle + buttons wala dialog	"Order Successful", "Top Up Successful" popups.

core/constants/

File	Kya aayega	Use
`app_constants.dart`	Tax rate, shipping types ki list, quick-offer amounts, spacing values, collection names (`'cars'`, `'orders'`)	Magic numbers/strings code mein bikhre na hon.

core/utils/

File	Kya aayega	Use
`formatters.dart`	`formatPrice(170000)` -> `$170,000`, date format, masked card number (`**** 4679`)	Formatting logic ek jagah. `intl` package yahan use hota hai.
`validators.dart` (optional)	Email, offer amount, PIN validation functions	Input check karne ke pure functions.

lib/data/

Poora data layer. Is folder mein Flutter ka UI code (widgets) kabhi nahi aata.

data/models/

Plain Dart classes. Har model mein `fromMap()` (Firestore se padhne ke liye) aur `toMap()` (Firestore mein likhne ke liye) hona chahiye.

File	Kya aayega	Use
`car.dart`	`id`, `name`, `brand`, `price`, `imageUrls`, `specs`, `sellerId`	Home aur car detail mein dikhne wali car.
`offer.dart`	`id`, `carId`, `buyerId`, `amount`, `status` (enum: processing/accepted/rejected), `createdAt`	Offer flow ka data.
`order.dart`	`id`, `carId`, `address`, `shippingType`, `amount`, `shipping`, `tax`, `status` (enum: verified/packing/inTransit/delivered), getter `total`	Checkout se tracking tak ka data.
`chat_message.dart`	`id`, `text`, `senderId`, `time`, `images`	Chat bubbles.
`wallet_transaction.dart`	`id`, `title`, `amount`, `date`, `isTopUp`	E-wallet history list.
`review.dart`	`id`, `userName`, `rating`, `comment`, `createdAt`, `likes`	Reviews screen aur Leave a Review sheet.
`address.dart`	`id`, `label` (Home/Office...), `fullAddress`, `isDefault`	Shipping address selection.

data/repositories/

Har repository ka pehle **abstract class** banao, phir uske do implementations: `MockXRepository` (fake data) aur `FirebaseXRepository` (real).

File	Kya aayega	Use
`car_repository.dart`	`getCars()`, `getCarById()`, `searchCars()`	Home aur detail ka data source.
`offer_repository.dart`	`submitOffer()`, `watchOfferStatus()`	Offer bhejna aur uska status follow karna.
`order_repository.dart`	`createOrder()`, `getOrders()`, `getOrderById()`	Checkout, My Orders, Track Order.
`chat_repository.dart`	`getChats()`, `watchMessages()`, `sendMessage()`	Inbox aur chat (Firestore stream ke saath).
`wallet_repository.dart`	`getBalance()`, `getTransactions()`, `topUp()`, `deduct()`	E-wallet aur wallet se payment.
`review_repository.dart`	`getReviews()`, `addReview()`	Reviews screen aur review sheet.

data/services/

Backend ke saath seedha baat karne wala code. Sirf Repositories inhe call karti hain.

File	Kya aayega	Use
`firestore_service.dart`	Generic helpers: `getCollection()`, `getDoc()`, `setDoc()`, `streamCollection()`	Firestore ke calls ek jagah, taaki repositories saaf rahein.
`auth_service.dart`	`signIn()`, `register()`, `signOut()`, `currentUser`	Firebase Auth wrapper.
`storage_service.dart` (baad mein)	Image upload, URL return	Chat images aur car photos ke liye Firebase Storage.

lib/features/

Har feature ke andar teen folders hote hain: `view/`, `viewmodel/`, `widgets/`.

- **view/**: screen files (poora page).
- **viewmodel/**: us feature ka **ek** ViewModel (screen-wise nahi, feature-wise).
- **widgets/**: sirf is feature mein use hone wale chhote widgets. Agar dusre feature mein bhi chahiye toh `core/widgets/` mein le jao.

features/offer/

Kaam: buyer car pe price offer karta hai aur seller ka jawab aata hai.

File	Kya aayega	Use
`view/make_offer_screen.dart`	Amount display, quick-amount chips, `NumPad`, "Send Offer" button	Offer amount enter karna.
`view/offer_processing_screen.dart`	Loading illustration, "Your offer is being processed" text	Seller ke response ka wait.
`view/offer_accepted_screen.dart`	Success illustration, "Proceed to Checkout" button	Accept hone pe checkout shuru.
`view/offer_rejected_screen.dart`	Sad illustration, "Make an Offer Again" aur "Back to Home"	Reject hone pe retry.
`viewmodel/offer_viewmodel.dart`	`amount`, `addDigit()`, `backspace()`, `sendOffer()`, offer status state	Offer ka saara logic aur state, chaaron screens ke liye ek hi.
`widgets/amount_display.dart`	Bada price text box	Make offer screen ka hissa.
`widgets/quick_amount_chips.dart`	$170,000 jaise suggestion chips	Tap karke amount fill.

features/checkout/

Kaam: accepted offer ka address, shipping aur total tay karna.

File	Kya aayega	Use
`view/checkout_screen.dart`	Selected address, car summary card, shipping type row, price breakdown	Checkout ka main page.
`view/shipping_address_screen.dart`	Address ki radio list, "Add New Address", Apply	Address chunna.
`view/choose_shipping_screen.dart`	Truck/Train/Ship/Plane options with price	Shipping method chunna.
`view/review_summary_screen.dart`	Final amount + shipping + tax = total, address, payment method, "Confirm Payment"	Payment se pehle last check.
`viewmodel/checkout_viewmodel.dart`	`selectedAddress`, `selectedShipping`, getter `total`, `addAddress()`	**Total ka calculation yahin hota hai, UI mein nahi.**
`widgets/address_tile.dart`	Ek address ka radio row	Address list mein repeat hota hai.
`widgets/shipping_option_tile.dart`	Icon + name + price + radio	Shipping list mein repeat hota hai.
`widgets/price_breakdown.dart`	Amount / shipping / tax / total rows	Checkout aur review summary dono mein.

features/payment/

Kaam: payment method chun kar PIN se confirm karna.

File	Kya aayega	Use
`view/payment_methods_screen.dart`	Wallet, PayPal, Google Pay, Apple Pay, Card ki list	Payment method chunna.

File	Kya aayega	Use
`view/enter_pin_screen.dart`	`PinDots` + `NumPad`, "Enter your PIN"	Payment confirm karna (top up mein bhi reuse).
`view/order_success_screen.dart`	Success popup, "View Order" aur "View E-Receipt"	Order ho gaya, ab orders pe jao.
`viewmodel/payment_viewmodel.dart`	`selectedMethod`, `pin`, `verifyPin()`, `pay()`	Payment ka logic. Wallet se pay hone pe `WalletRepository.deduct()` call karta hai.
`widgets/payment_method_tile.dart`	Logo + name + radio	Methods list ka ek row.

features/orders/

Kaam: orders dikhana, track karna, review dena.

File	Kya aayega	Use
`view/my_orders_screen.dart`	Active / Completed tabs, order cards, empty state	Saare orders ki list.
`view/track_order_screen.dart`	Car card + vertical status timeline	Order kahan tak pahuncha.
`view/leave_review_sheet.dart`	Bottom sheet: star rating + comment + Submit	Completed order pe review.
`viewmodel/orders_viewmodel.dart`	`activeOrders`, `completedOrders`, `loadOrders()`, `submitReview()`	Orders ka state aur filtering logic.
`widgets/order_card.dart`	Car image, naam, price, "Track Order" button	List ka ek card.
`widgets/status_timeline.dart`	Dots + lines + timestamp wali timeline	Track screen ka main widget.

features/reviews/

Kaam: ek store/car ke saare reviews dekhna.

File	Kya aayega	Use
`view/reviews_screen.dart`	Rating summary (4.8), star filter chips, review list	Reviews padhna.
`viewmodel/reviews_viewmodel.dart`	`reviews`, `selectedStarFilter`, `filteredReviews`, `loadReviews()`	Filter logic.
`widgets/review_tile.dart`	Avatar, naam, stars, comment, likes	List ka ek review.
`widgets/star_filter_bar.dart`	All / 5 / 4 / 3... chips	Filter select karna.

features/chat/

Kaam: dealers ke saath chat aur call.

File	Kya aayega	Use
`view/inbox_screen.dart`	Chats / Calls tabs, conversation list	Saari conversations.
`view/chat_screen.dart`	Message bubbles, input bar, image attach, typing state	Ek dealer se chat.
`view/call_screen.dart`	Dealer logo/naam, timer, mute/speaker/end buttons	Call UI (abhi sirf design, real calling optional).
`viewmodel/chat_viewmodel.dart`	`conversations`, `messages`, `sendMessage()`, `attachImage()`	Chat ka state, messages ka stream subscription.
`widgets/message_bubble.dart`	Sent/received bubble with time	Chat list ka ek message.

File	Kya aayega	Use
`widgets/chat_input_bar.dart`	Text field + attach + mic/send button	Chat ke neeche ka input.
`widgets/conversation_tile.dart`	Avatar, naam, last message, unread badge	Inbox ka ek row.

features/wallet/

Kaam: e-wallet balance, top up, history.

File	Kya aayega	Use
`view/wallet_screen.dart`	Balance card, "Top Up" button, transaction history	Wallet ka main page.
`view/top_up_screen.dart`	Amount display, `NumPad`, payment method chunna	Paise add karna.
`view/top_up_success_screen.dart`	Success popup, "View E-Receipt"	Top up complete.
`viewmodel/wallet_viewmodel.dart`	`balance`, `transactions`, `topUp()`, `loadWallet()`	Wallet ka state aur logic.
`widgets/balance_card.dart`	Card design with balance	Wallet screen ka top.
`widgets/transaction_tile.dart`	Icon, title, date, +/- amount	History ka ek row.

Naye features jo kit mein nahi hain

Roadmap ke Phase 2 mein ye banane padenge. Same MVVM pattern mein.

Folder	Files	Use
`features/home/`	`view/home_screen.dart`, `view/car_detail_screen.dart`, `viewmodel/home_viewmodel.dart`, `widgets/car_card.dart`	Car list, search, detail aur "Make an Offer" ka entry point.

Folder	Files	Use
`features/auth/`	`view/login_screen.dart`, `view/register_screen.dart`, `viewmodel/auth_viewmodel.dart`	Login/register, Firebase Auth ke saath.
`features/profile/`	`view/profile_screen.dart`, `viewmodel/profile_viewmodel.dart`	Bottom nav ka Profile tab, addresses aur PIN set karna.

test/ folder ka structure

`test/` mein `lib/` ka structure mirror karo.

File	Kya test hoga
`test/viewmodels/checkout_viewmodel_test.dart`	Shipping badalne pe total sahi update ho raha hai.
`test/viewmodels/offer_viewmodel_test.dart`	`addDigit` / `backspace` aur empty amount pe `sendOffer` fail.
`test/viewmodels/payment_viewmodel_test.dart`	Galat PIN reject, sahi PIN accept.
`test/widgets/num_pad_test.dart`	Key tap pe callback fire hota hai.

Mock repositories ki wajah se ye tests Firebase ke bina chal jaate hain.

Naming rules

- **Files:** `snake_case.dart` (`make_offer_screen.dart`).
- **Classes:** `PascalCase` (`MakeOfferScreen`).
- **Screens:** naam ke end mein `_screen`, bottom sheets ke end mein `_sheet`.
- **ViewModels:** `xyz_viewmodel.dart` -> class `XyzViewModel`.
- **Repositories:** abstract ka naam `OfferRepository`, implementations `MockOfferRepository` aur `FirebaseOfferRepository`.
- **Ek file, ek main class.** Do alag screens ek file mein mat daalo.

Nayi file kahan banani hai? (Quick check)

1. Sirf UI hai aur ek hi feature ka? -> `features/<feature>/view/` ya `widgets/`.
2. Do ya zyada features mein chahiye? -> `core/widgets/`.
3. State ya logic hai? -> `features/<feature>/viewmodel/`.
4. Data ka shape hai? -> `data/models/`.
5. Data kahan se aaye ye decide karna hai? -> `data/repositories/`.
6. Firebase/API ka seedha call hai? -> `data/services/`.
7. Colour, font, spacing? -> `core/theme/` ya `core/constants/`.